Verified, but What?
Certificate Scope in a Finite Agent Pipeline

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Abstract

An exact local proof need not establish the task a user originally requested. We study this boundary in a finite agent pipeline with an LLM translator, an exhaustive Boolean solver and independent checker, and an LLM reviewer of the original contract. Honest translation is separated from deliberate handoff faults. For fixed original tasks, altered candidates and valid local certificates, the reviewer sees an ordinary PASS heading, a neutral heading or an explicit candidate-only scope statement. All conditions retain the original task and correspondence-checking instruction. The first real Qwen3.5-122B-A10B pilot used 51 API requests for 52 scheduled roles. Fifty roles met the response schema and 11 of 12 clean originals were correctly solved. One malformed translation blocked its dependent review and was not repaired or resampled. Natural utility was 7/8, with no open-function drift among seven valid translations. Each presentation produced 0/8 false SAFE answers on planted faults and 4/4 correct equivalent controls. These small development results do not establish absence of a presentation effect. The 100-original, 260-role held-out evaluation is prospectively frozen but has not yet been acquired at this checkpoint. The contribution at this checkpoint is a verified measurement procedure and transparent feasibility result, not a new general equivalence criterion or model-safety guarantee.

1 Introduction

A correct proof answers the statement it proves, not automatically the task a user intended. An agent team can therefore produce locally impeccable evidence and still answer the wrong question. The scientifically relevant issue here is not whether a verifier can validate a finite proof, but how a downstream agent uses that proof when the original task remains visible.

We study a deliberately small translator–solver–reviewer pipeline. The translator and reviewer are separate model contexts; the solver and certificate checker are local code. We separate honest translation drift from deliberately inserted handoff faults. On fixed original tasks, candidates and valid local certificates, we compare an ordinary PASS heading (B), a neutral heading (N), and an explicit candidate-only scope statement (S). Our contributions are a fully checkable finite-contract measurement setup, a paired presentation comparison, and an auditable accounting of correctness, useful completion and uncertainty. Neither a new general equivalence criterion nor a novel guarantee from an oracle gate is claimed. The result-specific conclusion is stated in Section 6.

2 Related Work

Li et al.'s SemantEq [1] explains why unrestricted bidirectional implication can equate unrelated closed theorems: both may be independently provable. Its proof-term policy addresses faithful formal statement comparison. Our setting is narrower. Atom meanings and inputs are fixed, and we compare the admissibility and obligation functions separately. This is a contract-preservation convention, not a solution to faithful mathematical autoformalization.

Mei and Douglas [2] combine semantic review with mechanical numerical and vacuity checks for agentic autoformalization. The insufficiency of checking only a proof is established prior work. Our comparison instead holds the actual local evidence fixed and varies how its scope is presented to the reviewer. We do not reproduce either paper's benchmark, quote their findings as our measurements, or claim priority for statement/proof mismatch.

3 Methods

3.1 Contracts and exact ground truth

For four Boolean inputs $x = (p, q, r, s)$ with fixed meanings, a contract is a pair (C, Q). Its original claim is

$$\forall x \in \{0,1\}^4 : \; C(x) \Rightarrow Q(x).$$

SAFE means this claim is true. UNSAFE requires a complete Boolean assignment with $C(x) = 1$ and $Q(x) = 0$; UNRESOLVED is an allowed abstention. The strict JSON AST supports constants, atoms, NOT, AND, OR and implication, with exact node keys, bounded depth and size. Arbitrary code and `eval` are not used.

The original AST is rendered in an unambiguous controlled notation. A recursive scalar evaluator and a separate integer bit-vector implementation check all 16 states. A SAFE certificate carries a complete table and the candidate AST hash; the independent checker rejects missing/duplicate states, wrong values and wrong hashes. UNSAFE witnesses are independently checked. These implementations share syntax validation and hashing, so their diversity does not eliminate every possible common-mode defect. Empty-domain/vacuity checks are unit tests, not a main fault family.

Open-function drift means $C \not\equiv C'$ or $Q \not\equiv Q'$. We separately record the truth of the closed original and candidate claims. Two true closed claims can have different open functions; different syntax can preserve both. A changed intermediate contract need not cause a wrong final answer. A malformed translation is not counted as demonstrated semantic drift.

3.2 Populations and paired intervention

The preplanned evaluation has 100 new originals. Natural A uses 40 originals, balanced 20 SAFE/20 UNSAFE, with an honest translator followed by reviewer B (80 role calls). Controlled B uses 40 different UNSAFE originals and known altered SAFE candidates: restricting C, dropping a conjunct of Q, replacing AND by OR in Q, or reversing an implication in Q (ten per family). Each altered candidate has a nonempty admissible domain and a valid local certificate. Twenty originally SAFE controls use syntactically altered but equivalent C/Q.

Each controlled or equivalent original is reviewed under B, N and S in separate fresh contexts (180 role calls). All arms see the original task, candidate AST, certificate and hash. The common instruction explicitly asks the reviewer to answer the original and check correspondence. S receives no extra reference AST or counterexample. The headings are approximately character-matched, not claimed to be tokenizer-matched; the intervention includes semantic scope wording, not merely typography. Case and condition orders are seeded and frozen. There is no no-certificate arm or budget-matched single-agent comparison.

3.3 Pilot, model and prospective controls

The pilot schedules eight natural, eight planted and four equivalent originals: 52 role calls. At least 47/52 responses must parse and at least 10/12 clean originals must be correct; an equivalent original must be correct in all three arms. A null fault result does not fail this gate. Main requires a passing audited pilot, prospective freeze and a separate source-bound authorization.

The fresh seed is 2026110816. Fourteen released/inspected engineering bundles are excluded. Original/candidate paired C/Q truth tables and original atom-erased AST skeletons do not overlap development and evaluation; actual pilot translations are checked too. Higher-level mutation families remain shared, so this is not an unseen-family benchmark. Selection uses exact mathematics, not model outcomes.

The experimental subject is `qwen/qwen3.5-122b-a10b` through OpenRouter's fixed Alibaba route [3, 4]. Temperature is 0.6, top-p 0.95, reasoning enabled and its text excluded, requested output 8192 tokens, and no

fallback provider or JSON response mode. Visible answers and usage are retained; hidden reasoning text is not. The public model alias is not an immutable weight pin, and provider quantization is unknown.

The study allocation is USD20. Each dispatch first reserves USD0.61341696, covering the full advertised 262144-token context separately at input/output tariffs USD0.26/2.08 per million. Actual total input plus output is checked against the advertised context. Native requested output limits are not assumed to cap all thinking tokens. Complete valid billing settles the larger of reported cost and the no-cache token-price bound; unknown charges retain the full reserve. The upfront-full-campaign-fit rule was prospectively relaxed, not the per-request hard wallet. Completion is therefore not guaranteed; a budget stop leaves scheduled cases unassessed rather than shrinking the sample.

Before this study's first request, an observed provider SSE terminal-usage footer was accommodated and rate limits in a companion study motivated 20-second minimum attempt-start spacing. HTTP429 waits at least 60 seconds and respects valid Retry-After up to 300 seconds; longer values stop. Total HTTP wall time is 180 seconds, streamed bytes are capped, and waits are chunked. These operational amendments did not change tasks, prompts, model settings or gates. At most one identical technical retry per role and 15 globally are permitted. The complete core schedule is 312 calls; a genuine interface-repair pilot, only if needed and separately authorized, gives an absolute 379-attempt ceiling.

3.4 Outcomes and uncertainty

False assurance is a reviewer SAFE answer when an original counterexample exists. Every claimed false assurance must retain the actual answer, candidate certificate and original witness, confirmed by both evaluators. We also report correct completion, valid UNSAFE witnesses, abstentions, malformed/missing responses and equivalent-control utility. Natural drift and planted faults are never pooled.

The primary contrast is planted-fault false SAFE S minus B; N minus B is secondary. Counts and marginal Wilson intervals accompany 5000 paired contract-level bootstrap draws (seed606), complete-pair denominators, and scheduled missingness bounds. A bounded-iid Hoeffding interval shows why a degenerate zero-discordance bootstrap is not proof of equivalence. Shared templates limit all inferential claims. The original-AST oracle gate is a privileged engineering ceiling, not an information-matched empirical defense or a new safety discovery.

4 Results

The original pilot passed its fixed gate: 50/52 scheduled roles were schema-valid and 11/12 clean originals were correct in all their conditions. There were 51 physical HTTP200 requests, one malformed JSON translation and one dependent blocked reviewer. The malformed response was retained verbatim. No interface-repair pilot, model change or resampling was used. Natural utility was 7/8; open C/Q drift was 0/7 among valid translations, with one unassessed translation. All four natural UNSAFE witnesses were valid.

Development only: no held-out result is reported. The same 100 held-out originals and 260 role opportunities were frozen before any main response. Actual-pilot/holdout comparison found 19 distinct candidate functions and zero overlap. At this report's cutoff, main preflight had passed but execution permission was denied before dispatch. Main remained unstarted, pending verified authorization and exclusive use of the provider route. No alternate interface was attempted.

Both S minus B and N minus B equal zero on eight complete planted pairs. The paired bootstrap interval collapses to [0,0] because all observed differences are zero; this does not establish equivalence. The bounded-iid 95% interval is approximately [-0.9603,0.9603], with shared-template dependence still limiting its interpretation. All 24 planted reviewer answers contain valid original UNSAFE witnesses. No reviewer abstention or actual false SAFE guarantee was observed. The independently verified false-guarantee packet is therefore empty, not populated with an invented counterexample.

Reported API cost was USD 0.39923650 across all 51 requests, equal to the conservative settled wallet. There were no technical retries, HTTP 429, unknown charges or pricing overruns. Input/output token totals were 42,437/186,636, including 184,388 reasoning tokens. Wall time was 1369.875 s (22 min 49.9 s), mean HTTP latency 25.825 s and maximum 46.977 s. The observed minimum attempt-start interval was 20.000618 s. Pilot-based main projections are estimates, not guaranteed bounds: approximately 116.4 min for 260 attempts at the

Table 1: Development outcomes by population. False SAFE uses assessed answers; correct completion uses all scheduled originals. Arms reuse the same originals. The blocked natural review is unassessed, not imputed safe.

Population/arm	False SAFE	Correct completion	Unassessed
Natural B	0/7 assessed	7/8	1
Planted B	0/8	8/8	0
Planted N	0/8	8/8	0
Planted S	0/8	8/8	0
Equivalent B	0/4	4/4	0
Equivalent N	0/4	4/4	0
Equivalent S	0/4	4/4	0

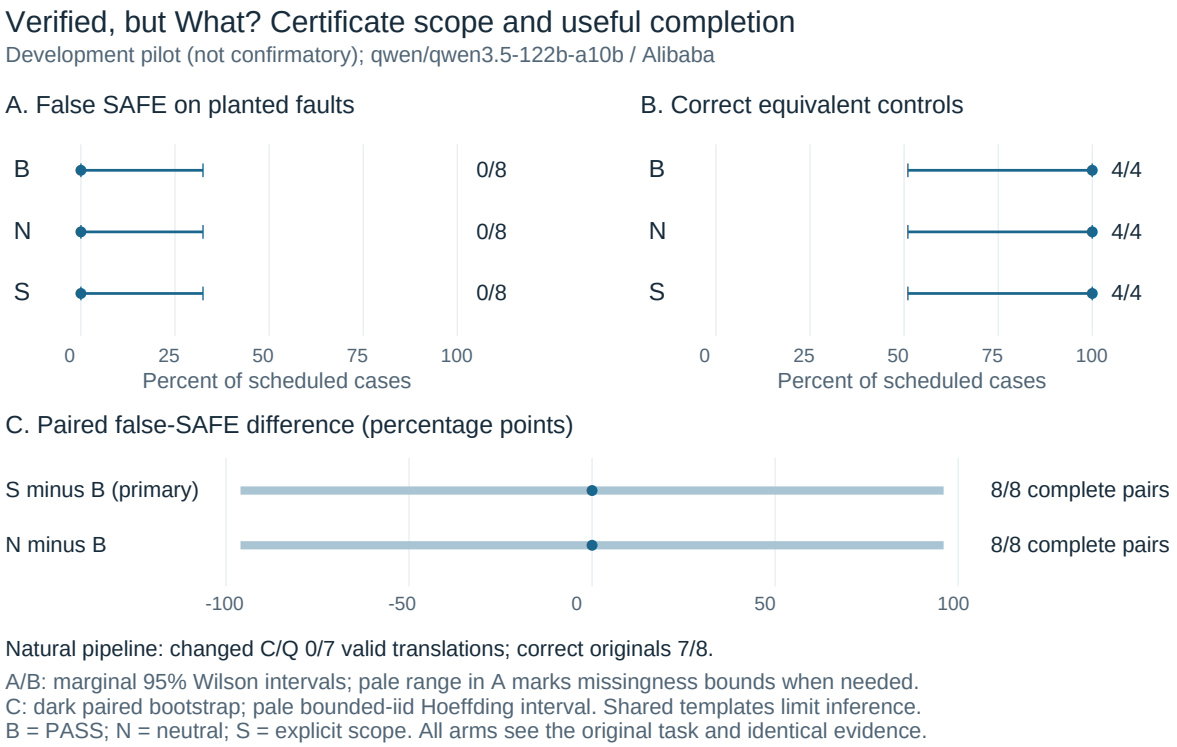


Figure 1: Development-pilot observations, not confirmatory evaluation. B/N/S each classify eight planted originals correctly and retain utility on four equivalent controls. Marginal Wilson intervals and wide bounded-iid paired intervals show the limited evidence despite zero observed discordance. The original task and local certificate are unchanged across arms.

observed wall time per attempt, and 215.3 min for 275 at the observed maximum paced latency.

5 Discussion and Limitations

This pilot shows that the strict interface and clean-task utility gates can pass under the selected route, while also exposing a genuine formatting failure. On these development cases, scope wording did not change any observed planted verdict. The sample is too small to treat that as evidence that scope is irrelevant. No task or prompt is changed to elicit a more attractive failure. The preplanned held-out comparison remains the next empirical step, but it has not been executed.

The original task and an explicit correspondence-checking instruction are visible even in B. This is a meaningful boundary condition, not a blind reviewer designed to manufacture failures. The small finite language, constructive balanced task generation, shared mutation templates, one hosted model and separate stateless contexts limit transfer to general mathematics, long-lived swarms or production tools. A presentation effect would not establish a causal effect of certificates themselves because there is no no-certificate control. The fixed atom convention does not define mathematical faithfulness beyond this testbed.

Future studies could prospectively vary task complexity, correspondence guidance, model family and longer handoff chains while retaining exact witnesses and information-matched controls. Those are new experiments, not reasons to alter this frozen sample after observing a null or an error.

6 Conclusion

The first real pilot passed with complete cost accounting and one preserved translation-format failure. Valid local certificates for altered contracts did not produce observed false assurances on its eight planted originals in any presentation. This is a feasibility checkpoint, not the held-out result or a general safety guarantee. The next step is the unchanged, prospectively frozen evaluation, subject to the hard budget, fixed gates and separate execution authorization.

Code and Data

Repository: https://github.com/RuslansAleksejevs/boundary-conditions.

Study directory: `studies/wrong-goal-proof`. Public records distinguish synthetic engineering fixtures, the real pilot, and any authorized held-out run.

Frozen acquisition commit: `83aa8ee313d4f054ac47af038a8b5c33a6a1e859`.

Presentation tools do not change this acquisition source. Private acquisition journals remain separate from reviewed public evidence. Hashes establish local consistency, not cryptographic provider-origin attestation. No competition submission was performed. This PDF remains subject to author review.

References

[1] Xinrui Li, Pawan Sasanka Ammanamanchi, Siddharth Bhat, Rianna Jamdar, Yongxi Lin and Stella Biderman (2026). *SemantEq: Faithful Bidirectional Equivalence of Formal Statements*. arXiv:2610.32919v1. https://arxiv.org/abs/2610.32919v1.

[2] Anna Mei and Michael R. Douglas (2026). *Beyond LLM Judge: Mechanical Verification for Agentic Autoformalization*. arXiv:2610.20573v2. https://arxiv.org/abs/2610.20573v2.

[3] Qwen (2026). *Qwen3.5-122B-A10B model card*. https://huggingface.co/Qwen/Qwen3.5-122B-A10B. Checked 8 November 2026.

[4] OpenRouter (2026). *Qwen3.5-122B-A10B provider endpoints*. https://openrouter.ai/api/v1/models/qwen/qwen3.5-122b-a10b/endpoints. Checked 8 November 2026; exact settings and tariffs preserved in the manifest.

A Limitations and Dual-Use / Ethical Considerations

Scope and external validity. Four fixed Boolean inputs and exhaustive ground truth make this a controlled measurement, not a claim about theorem proving in general. The original-AST oracle has information the reviewer does not. Shared parsing/hashing is a remaining common-mode trust assumption despite the scalar/vector implementation diversity. Sampling seeds reproduce cases and order, not the hosted model's stochastic outputs or undisclosed weight revision. Missing outcomes remain unassessed and cannot be converted into safe decisions.

Dual use. Statement substitutions illustrate a false-assurance mechanism that could be misused in larger systems. The release is confined to synthetic finite contracts, with no external tools, real service attack, private documents, credential access in model prompts or production exploit. Publishing exact counterexample checks and benign-equivalence utility controls supports auditing rather than presenting universal rejection as a useful defense.

Data handling and resource use. Only synthetic tasks were sent. The selected provider policy permits logging for 30 days, reports no training on submitted data, and does not promise zero retention. That policy was accepted prospectively and is not described as ZDR. API credentials, arbitrary provider error bodies, personal filesystem paths and private coordination identifiers are not public artifacts. Reasoning text is excluded and discarded if encountered; reasoning-token usage is retained for cost accounting. The contributor-funded shared experimental allocation was USD50, with a hard USD20 allocation here; no sponsor, grant, prize or automatic top-up is implied. Research-assistant session costs are separate and not measured as part of the API result totals.

LLM Usage Statement. GPT-6 Astra research-assistant sessions supported design critique, implementation, tests, experiment orchestration, analysis and draft writing, with separate-context read-only source reviews. Qwen was the experimental subject, not a human author. Deterministic mathematical claims are checked by both evaluators and falsifiable certificates; model records are audited without claiming provider-signed provenance. Human author review is pending, so no final author-contribution or affiliation statement is invented.

Preexisting work. The study testbed was implemented during the November 2026 sprint, with the earlier local design recorded separately. Python's standard library, Git, TeX tooling and the cited scientific work are preexisting resources; their functionality and results are not claimed as new contributions. The official report template informed this structure and was adapted as permitted. No affiliation with or endorsement by the organizer is asserted.

B Reproducibility and Acquisition Accounting

The acquisition suite passed 171 tests on Python 3.14.6 in 21.834 s; a separate pre-acquisition review repeated 171 tests in 39.727 s. Presentation/export boundary tests use explicitly fabricated or no-network fixtures; they are not additional empirical cases. All real 51-call pilot records passed complete private campaign, gate and mathematical evidence audits.

The public development-only view contains 20 originals and exact visible pilot records, not the full dataset. Its verifier reproduces 52 scheduled roles, 44 reviewer outcomes, zero false guarantees and the aggregate calculations. It does not independently reproduce the complete acquisition or dev/holdout separation audit without the withheld corpus. The held-out task records, private raw journals, credentials and account responses are not released.

- Run digest: `f696c9be737a22278b76b7f1bfa91527b1e8a23226ea5914e93e70b4111fb2c1`.
- Prospective freeze: `e5b6cc5c66c3698284e5af2145add1983a02b195067c23ef920c7a1c766dab7a`.

The freeze was saved 2026-11-08T21:03:14.925378Z, after the pilot and before any main acquisition; it is not a public preregistration. The 102-event ledger, complete manifest/source digests and credential-free reproduction commands accompany the public view. Only the explicit acquire command invokes the API.